



CHALLENGE 15 / 20

THE LATERAL MOVEMENT MAZE

ATTACK-PATH INFERENCE UNDER PARTIAL VISIBILITY

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

A state utility runs 6,400 hosts across 70 sites, including plant networks and substations, three directory forests and a contractor VPN. Only 31% of hosts have usable logging. The rest are legacy, embedded or Linux systems, or have logging disabled for performance. An attacker reached a domain controller and a historian.

Only nine observations exist over four months: a phishing click, two odd service creations, an anomalous Kerberos ticket and a VPN session among them. The attacker used legitimate administration tools, three service accounts and shared jump hosts used by hundreds of administrators. Firewalls log denied connections but not permitted ones. Network documentation is 40% out of date.

The utility cannot rebuild 6,400 hosts. It must know which to isolate and where the attacker may still be.

2 | PROBLEM STATEMENT

Develop an **Attack Path Inference System** that infers the most probable attack paths and the set of compromised hosts from sparse observations and an uncertain network topology, reconstructing the real topology from partial evidence as it goes.

The system must recommend a minimal containment set with quantified residual risk, rank which unobserved hosts to inspect to reduce uncertainty the most, and update as new evidence arrives.

Hard constraints

- Sixty-nine percent of the estate is invisible and the documented topology is wrong.
- Legitimate administrator behaviour hides the attacker's.
- The system must scale to thousands of hosts and very many candidate paths.
- Containment advice must carry a measured residual risk.
- The attacker adapts to avoid monitored hosts.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Topology Inference	Ability to infer the real network from partial and wrong documentation.
02	Sparse Observation Fusion	Ability to combine a few observations into path evidence.
03	Path Hypothesis Ranking	Ability to rank very many candidate attack paths.
04	Compromised-Host Estimation	Ability to estimate which hosts are compromised.
05	Containment Optimisation	Ability to propose a minimal containment set with residual risk.
06	Next-Inspection Selection	Ability to choose which hosts to examine to cut uncertainty most.
07	Sequential Evidence Updating	Ability to update conclusions as evidence arrives.
08	Adaptive-Adversary Handling	Ability to cope with an adversary avoiding monitored hosts.

The system must clearly distinguish between:

OBSERVED	INFERRED HOP	LIKELY PATH	ALTERNATIVE PATH	UNKNOWN
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A hop that was never observed must not be presented as observed.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated multi-site estate with wrong documentation and partial logging
02	Simulated adaptive attacker
03	Inferred topology
04	Ranked attack paths
05	Compromised-host estimate
06	Containment set with residual risk
07	Evidence-driven update after new observations

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.